

Gabriel Thien

Graduate Engineer (Embedded Software) @ Cochlear Ltd.
B. Engineering (Mechatronics) / B. Science (Computer Science)
Graduate @ UNSW

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[LinkedIn](#)

[GitHub Repositories](#)

QUALITIES

*Strong collaboration, teamwork,
leadership skills*

*Motivated, self-managing with
workload,
capable of working independently*

Proactive, forward thinking

*Strong communication,
presentation,
networking skills*

*Outgoing, friendly, approachable
personality*

SUMMARY

Embedded Systems / Mechatronics engineer with demonstrated experience in developing electro-mechanical systems, firmware and electronics. Motivated and self-initiated with a strong technical portfolio to match ranging R&D, prototyping, electronics, firmware and software tooling. Open to opportunities in mechatronics & embedded systems.

EDUCATION

UNSW Sydney – (2021 ~ 2025)

Degree: *B. Engineering (Mechatronics)/B. Science (Computer Science)*

EXPERIENCE

Graduate Engineer (Embedded Software)

Cochlear – (Feb 2026 – Present)

- Rotating through various departments, understanding and contributing and solving distributed problems in daily operations
- Solving multifaceted business problems with applied AI, software and hardware solutions, increasing automation and efficiency

Student Firmware Engineer

Cochlear – (Nov 2024 – Nov 2025)

- Developed a stack analyser that enabled optimised memory usage in constrained ASICs
- Helped reduce memory-related issues during product development, saving developers time
- Involved architectural thinking for firmware platforms and infrastructure projects

Undergraduate Mechatronics Engineer

Asiga 3D Printers – (March 2023 – Nov 2024)

- Developed PCB electronics, 3D printed jigs & tooling hardware that improved quality control
- Created device drivers and RTOS firmware for a variety of verification and testing platforms

Machine Technician

Archie Brothers Alexandria – (June 2022 – October 2022)

- Improved arcade machine up time with maintenance, fault diagnosis and repairs.

PROJECTS

Custom IoT Automation Development Platform

- Developed modular nodes housing microcontrollers, screens, motors, communication modules

Competitive Robotics @ RoboCup Junior Australia

- Engineered autonomous competitive robots with various sensors and interactive mechanisms
- Involved strategizing around constraints, optimising for performance

TECHNICAL SKILLS

*C, C#, C++, Python, Java, SQL
Object Oriented Programming
HTTP, JUnit, blackbox testing*

Agile Development, Git, Postman

*Linux, PLC programming, Qt software
development*

*Embedded software, FreeRTOS, MCU
application development in C/C++,
Arduino, Espressif, STM32
frameworks*

*PCB design, electronics circuitry
design, TH & SMD soldering, testing &
debugging*

*3D CAD, 3D printing, laser cutting,
hands on workshop skills*

*PC building & installation, desktop
support*

HOBBIES & INTERESTS

Bouldering, rock climbing, hiking

Mahjong